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1. ABC and DCB are similar

$$\therefore \frac{AB}{AC} = \frac{DC}{DB} \Rightarrow AB \times DB = DC \times AC$$

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2. Let (a, b) be any point in the plane

Distance from origin of (a, b) is (0, 0)

$$\begin{aligned} \text{Distance formula} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(a - 0)^2 + (b - 0)^2} \\ &= \sqrt{a^2 + b^2} \end{aligned}$$

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3. In $\triangle ABC$ and $\triangle DAC$

$$\angle BAC = \angle ADC - (\text{Given})$$

$$\angle ACB = \angle ACD - (\text{common})$$

$$\therefore \triangle ABC \sim \triangle DAC - \text{by AA similarity}$$

$$\Rightarrow \frac{AC}{BC} = \frac{DC}{AC}$$

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$$4. \quad x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2} \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

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$$5. \quad \text{To find} = \frac{\text{ar}(\triangle DEF)}{\text{ar}(\triangle ABC)}$$

$$\therefore FC = \frac{1}{2} BC$$

$$EC = \frac{1}{2} AC$$

$$EF = \frac{1}{2} AB$$

In $\triangle ABC$ and $\triangle EFD$.

$$\frac{AB}{EF} = \frac{BC}{FD} = \frac{AC}{ED} = 2 \quad \dots(1)$$

$\triangle ABC \sim \triangle EFD$...(sss similarity)

$$\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle EFD)} = \left(\frac{AB}{EF}\right)^2 = 4$$

(from i)

$$\frac{\text{ar}(\triangle EFD)}{\text{ar}(\triangle ABC)} = \frac{1}{4}$$

$\therefore$ The area is 1 : 4

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$$A(-3, 2) \quad B(6, 1)$$

6. Given - A $\begin{matrix} x_1 & y_1 \end{matrix}$ $\begin{matrix} x_2 & y_2 \end{matrix}$

Let the coordinate of points (o, y) divide the ratio m : n

$$(o, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$= \left(\frac{6m-3n}{m+n} + \frac{m+2n}{m+n} \right) = (o, y)$$

$$\therefore \frac{6m-3n}{m+n} = 0$$

$$\therefore 6m = 3n \quad \text{ie} = \frac{m}{n} = \frac{1}{2}$$

Ratio = 1 : 2

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7. In $\triangle ADE$ & $\triangle DABC$

$$\angle A = \angle A \quad (\text{common})$$

$$\angle EDA = \angle CBA \quad (\text{corresponding angles})$$

Hence, by AA similarity,

$\triangle ADE$ is similar to $\triangle ABC$

Similarity of Triangles-

$$\frac{AD}{AB} = \frac{DE}{BC} = \frac{AE}{AC}$$

$$\frac{AD}{AD+BD} = \frac{AE}{AE+EC}$$

$$\frac{AD+BD}{AD} = \frac{AE+EC}{AE}$$

$$1 + \frac{BD}{AD} = 1 + \frac{EC}{AE}$$

$$\frac{BD}{AD} = \frac{EC}{AE}$$

Substituting

$$\frac{x+4}{2x+7} = \frac{2x-1}{3x}$$

$$\frac{BD}{AD} = \frac{EC}{AE}$$

$$\frac{x+3}{x+4} = \frac{x+1}{2x-1}$$

$$2x^2 + 5x - 3 = x^2 + 5x + 4$$

$$x^2 = 7$$

$$x = \sqrt{7}$$

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9. $AD^2 + AB^2 + BD^2$ ($\triangle ABD$ is a $\angle 90^\circ$ at B)

$$AD^2 = AB^2 + \left(\frac{BC}{2}\right)^2 \quad [\because BD = DC]$$

$$AD^2 = AB^2 + \frac{1}{4} BC^2 \quad \dots(i)$$

($\triangle BCE$ is $\angle 90^\circ$ at B)

$$\therefore CE^2 = BC^2 + BE^2$$

$$CE^2 = BC^2 + \left(\frac{AB}{2}\right)^2 \quad [\because BE = EA]$$

$$CE^2 = BC^2 + \frac{1}{4} AB^2 \quad \dots(ii)$$

Add (i) and (ii)

$$AD^2 + CE^2 = AB^2 + \frac{1}{4} BC^2 + BC^2 + \frac{1}{4} AB^2$$

$$\Rightarrow AD^2 + CE^2 = \frac{5}{4} (AB^2 + BC^2)$$

$$AD^2 + CE^2 = \frac{5}{4} (AC^2) \quad [\because ABC \text{ is right triangle}]$$

$$\therefore [AC^2 = AB^2 + BC^2]$$

$$\Rightarrow \left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{5}{4} \times 25$$

$$CE^2 = \frac{125}{4} - \frac{45}{4} = 20$$

$$CE = \sqrt{20} \text{ cm} = 2\sqrt{5} \text{ cm.}$$

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10. Area of $\triangle ABC$

$$= \frac{1}{2} [2(4 - 6) + 4(6 - 2) + 2(2 - 4)]$$

$$= \frac{1}{2} [2(-2) + 4(4) + 2(-2)] = 4 \text{ sq. units.}]$$

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